ClonoTrace: deterministic clonal inference and alternative receptor models for B-cell repertoires

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Abstract

B-cell receptor clonal inference groups sampled sequences into hypotheses of shared ancestry. Inferred partitions depend on sequence representation, annotations and clustering rules. ClonoTrace is a deterministic Rust framework for evaluating alternative receptor models while reproducing selected reference-compatible profiles. Its workflows include hierarchical and spectral clustering, paired-chain refinement, amino-acid analysis, reference-directed germline reconstruction and threshold sensitivity. Complete member partitions matched Change-O 1.3.5 and SCOPer 1.6.0 in all 37 and 1,006 eligible comparison cases, respectively; exact clone counts matched in every case, with Pearson r=1 and Spearman rho=1. Synthetic examples show how linkage changes analytical groups, known light-chain signatures refine heavy-chain grouping, and translation groups synonymous nucleotide variants while retaining amino-acid distinctions. Germline and deletion analyses expose annotation policies and threshold sensitivity rather than ancestry or statistical confidence. For a 512-row spectral workflow, a storage and traversal optimization reduced one-worker median runtime from 5.03 to 1.15 seconds, or 4.39-fold faster, without changing results. Dense execution showed no meaningful worker scaling. The framework makes receptor-model choices explicit and reproducible; the evidence is computational and host-specific, not biological clone truth or superiority.

Introduction

B cells descended from a common receptor rearrangement can accumulate sequence differences through somatic hypermutation. Repertoire studies group sampled receptor sequences into inferred clones to study diversity, expansion and diversification. Because sequence groups depend on the evidence and model used to define them, an inferred clone is an analytical hypothesis of common ancestry rather than a direct observation. Junction similarity, V/J annotations and heavy-light pairing provide distinct information for this inference [1-3].

Established tools provide complementary analytical models. Change-O supports gene-aware clonal assignment [1], while SCOPer includes identity, hierarchical and spectral approaches [2]. AIRR representations help preserve annotated receptor information across tools [4]. Comparing analyses therefore requires separating the selected receptor model from method-specific grouping and input policies. Software concordance addresses continuity with an established workflow; alternative models address how a different analytical choice changes the inferred groups.

ClonoTrace addresses this challenge through a unified, deterministic framework for alternative receptor models alongside selected reference-compatible profiles. Its central contribution is to make linkage, graph resolution, chain pairing, sequence representation, reference annotation and threshold sensitivity explicit and auditable. Compatibility with selected Change-O and SCOPer workflows provides a basis for comparison without making software replication the main scientific objective. We

examine the analytical consequences of these choices and distinguish computational agreement from biological interpretation.

Methods

Receptor data representation and input partitioning

Analyses preserve donor and sample boundaries and use an explicit subset of AIRR-style annotated records. Bulk clonal inference analyzes IGH junctions; paired-chain and germline workflows also support IGK and IGL. Extended bulk models use single V/J genes and equal analyzed lengths.

Reference-compatible baseline workflows

Change-O 1.3.5 `DefineClones` is restricted to gene resolution, normalized nucleotide Hamming distance, average symmetrization, single linkage and maxmiss=0. Thresholds are explicit and inclusive; set/first call handling follows the frozen native profile. The SCOPer 1.6.0 `identicalClones` profile uses nucleotide identity, cdr3=FALSE, mod3=FALSE and max_n=0, with the specified first/set call action. These profiles do not cover every mode of either package.

Each baseline workflow is evaluated directly against complete member partitions generated by independent, unmodified reference software executions.

Alternative receptor models and extended workflows

Hierarchical analysis supports single, average and complete linkage. Spectral analysis uses Gaussian affinities and an explicit number of clusters per positive-edge component. Paired inference forms heavy-chain groups and refines complete pairs by exact known light signatures. Amino-acid analyses use direct canonical residues or standard-code translation with an explicit frame. Germline reconstruction projects supplied references through supplied annotations. Threshold assessment repeats a nearest-neighbor gap estimate after deleting each distinct junction. Full parameter, numerical and input policies are provided in Supplementary Methods and the accompanying user manual (see Software and data availability).

Hierarchy is row-weighted for average linkage. Spectral analysis uses a zero-diagonal Gaussian affinity, positive-edge components, a normalized Laplacian, deterministic cyclic Jacobi decomposition, canonical repeated-eigenspace bases and deterministic Lloyd assignment. The fixed limits include 100 Jacobi sweeps, off-diagonal tolerance 1e-13, eigen residual bound 1e-9, and at most 512 rows per dense work unit.

Evaluation corpus, concordance metrics, and performance benchmarking

Complete member partitions, independent of numeric labels, were the formal software-concordance outcome. The eligible descriptive count subset comprised 37 Change-O cases, with 23 unique inputs and 37 input/action/threshold combinations, and 1,006 SCOPer cases, with 799 unique inputs and 990 input/action/threshold combinations. Four combined Change-O probes were excluded intact from the count plot without splitting or summing them. Pearson r used clone counts; Spearman rho used average ranks for ties. Stored reruns remain execution cases, not independent biological samples. Supplementary Methods gives the full eligibility rules.

Extended methods were assessed using independent mathematical or policy expectations, synthetic receptor examples and two prepared public B-cell receptor tables. Illustrative sequences are deliberately short and do not supply biological lineage truth. Historical runtime comparisons used five sequential repetitions per configuration on a macOS ARM64 Mac16,11 with 24 GiB RAM. Wall time included process startup; builds were excluded. Figure whiskers show observed minima and maxima.

Results

ClonoTrace establishes a unified framework for comparative clonal inference

Inferred B-cell clones represent computational models rather than direct biological observations. Different analytical questions retain different kinds of receptor information. ClonoTrace brings selected reference-compatible profiles and alternative receptor models into one deterministic workflow (Figure 1).

The framework preserves donor and sample boundaries while exposing hierarchical linkage, spectral resolution, known chain pairing, amino-acid representation, reference annotations and threshold sensitivity as analysis choices. Source observations, exclusions and results remain traceable. These choices provide a coherent basis for comparing models, while their partitions retain different analytical meanings.

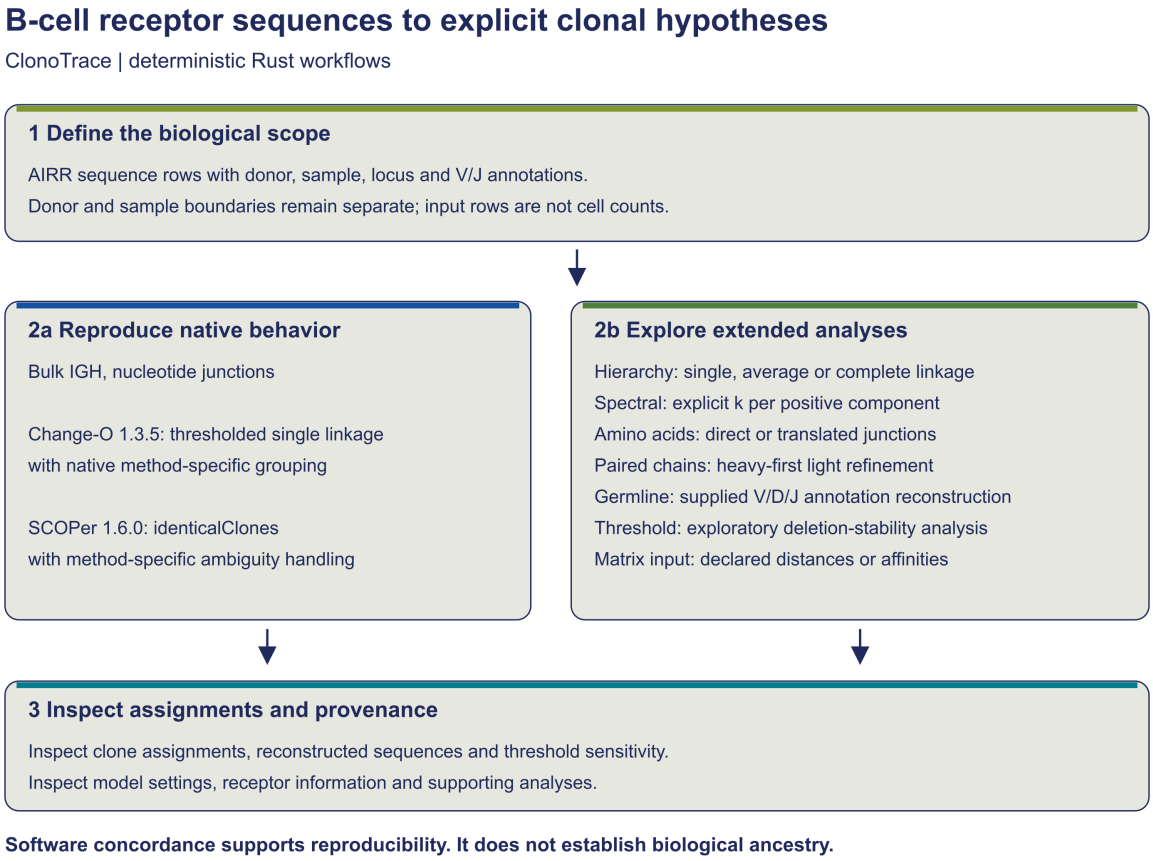


Figure 1. ClonoTrace workflow and scientific scope. Annotated B-cell receptor records remain separated by donor and sample. Selected Change-O and SCOPer profiles support reproducible assignment; extended analyses expose linkage, graph structure, chain pairing, amino-acid representation, reference annotations and threshold sensitivity. Source rows and analysis choices remain traceable. Germline reconstruction produces reference-derived sequences rather than clone assignments.

Exact partition agreement reproduces selected Change-O and SCOPer reference profiles

Establishing continuity with widely adopted clonal inference software is a prerequisite for comparative analysis. ClonoTrace was evaluated against independent, unmodified executions of Change-O 1.3.5 and SCOPer 1.6.0 across an extensive validation corpus. Complete member-partition equality served as the formal criterion of software concordance.

Across all 37 eligible Change-O cases and all 1,006 eligible SCOPer cases, ClonoTrace matched the complete member partitions exactly (37/37 and 1,006/1,006; Figure 2A-B). Clone counts also matched in every plotted case, with Pearson r=1 and Spearman rho=1 in both panels. Four combined Change-O probes were excluded intact from this descriptive plot rather than converted into aggregate observations.

Clone-count agreement alone cannot establish partition equality: different assignments can contain the same number of groups. Complete member-partition comparison was therefore the formal

concordance test within the selected profiles. This establishes software agreement on the compared cases, not biological accuracy or common ancestry.

Complete member partitions also agreed on the prepared 2,246-row public table. Both tools retained 1,097 IGH rows, excluded 1,149 other rows and failed zero rows. Change-O-compatible assignment at threshold 0.15 produced 1,092 clones; SCOPer-compatible identity produced 1,095. Each matched its own reference. The different counts reflect different inference rules; historical command runtimes are reported in Supplementary Figure S1.

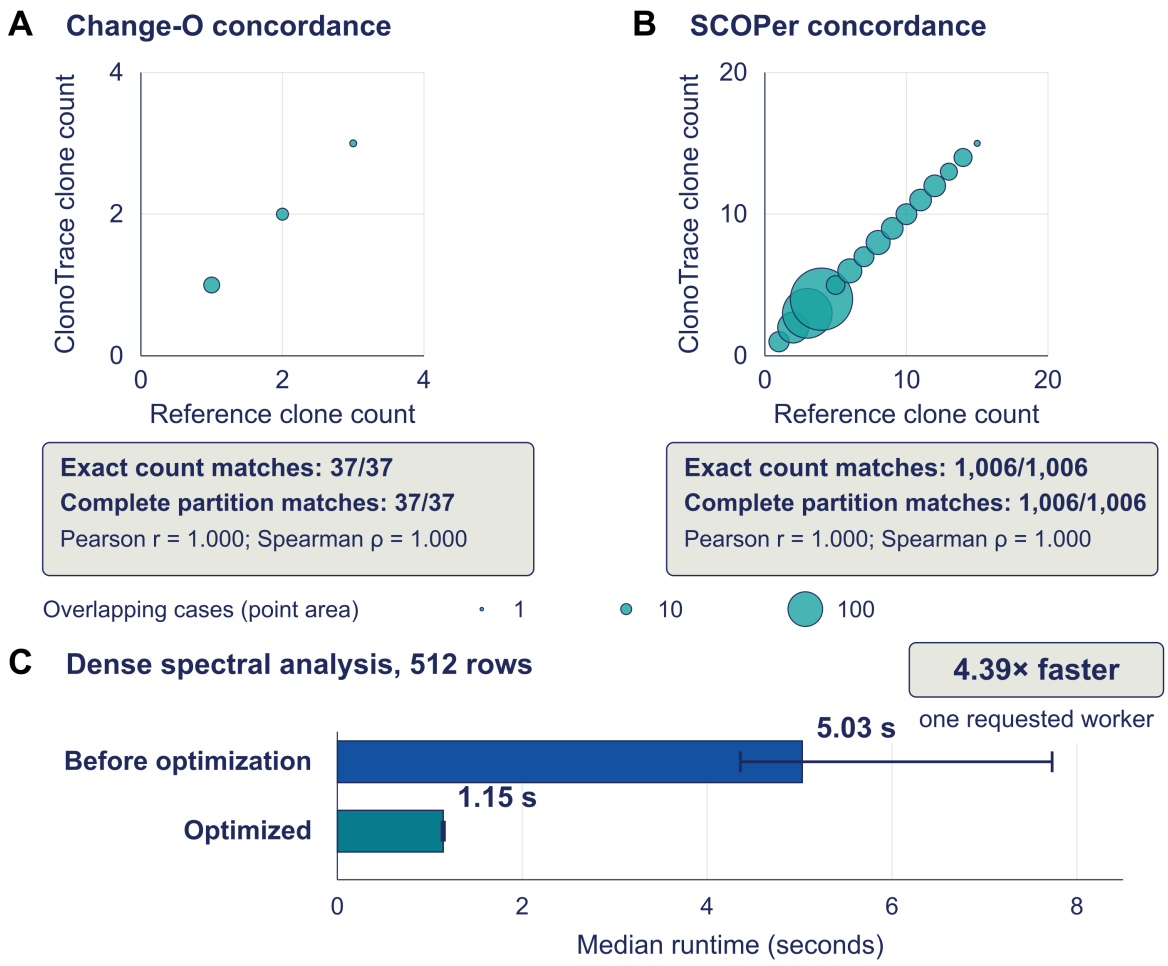


Figure 2. Software concordance and dense spectral runtime. A-B: ClonoTrace 1.0.0-rc.1 versus Change-O 1.3.5 (37 cases) and SCOPer 1.6.0 (1,006 cases). Square plots have equal x/y units; marker area indicates overlapping cases. Result boxes report exact count and complete partition matches, Pearson r and Spearman ρ. Correlation is descriptive; complete partition agreement is the formal concordance result, not evidence of biological accuracy. C: Solid horizontal bars show median runtime for 512-row spectral analysis before and after optimization, with five-run minimum-to-maximum whiskers: 5.03 to 1.15 s, 4.39× faster with one requested worker. Dense execution showed no meaningful worker scaling. Supporting methods give eligibility and exact measurements.

Clustering topology and spectral graph partitioning alter the inferred clonal structure

Single linkage permits grouping through chains of pairwise similarity.

In the synthetic AAAA, AAAC, AACC and ACCC example, normalized Hamming threshold 0.25 connects all four sequences. Complete linkage at the same threshold returns {a,b}|{c,d}; spectral analysis (sigma 0.25, k=2) gives the same partition under a different model (Figure 3A). Agreement here does not establish general equivalence between the models.

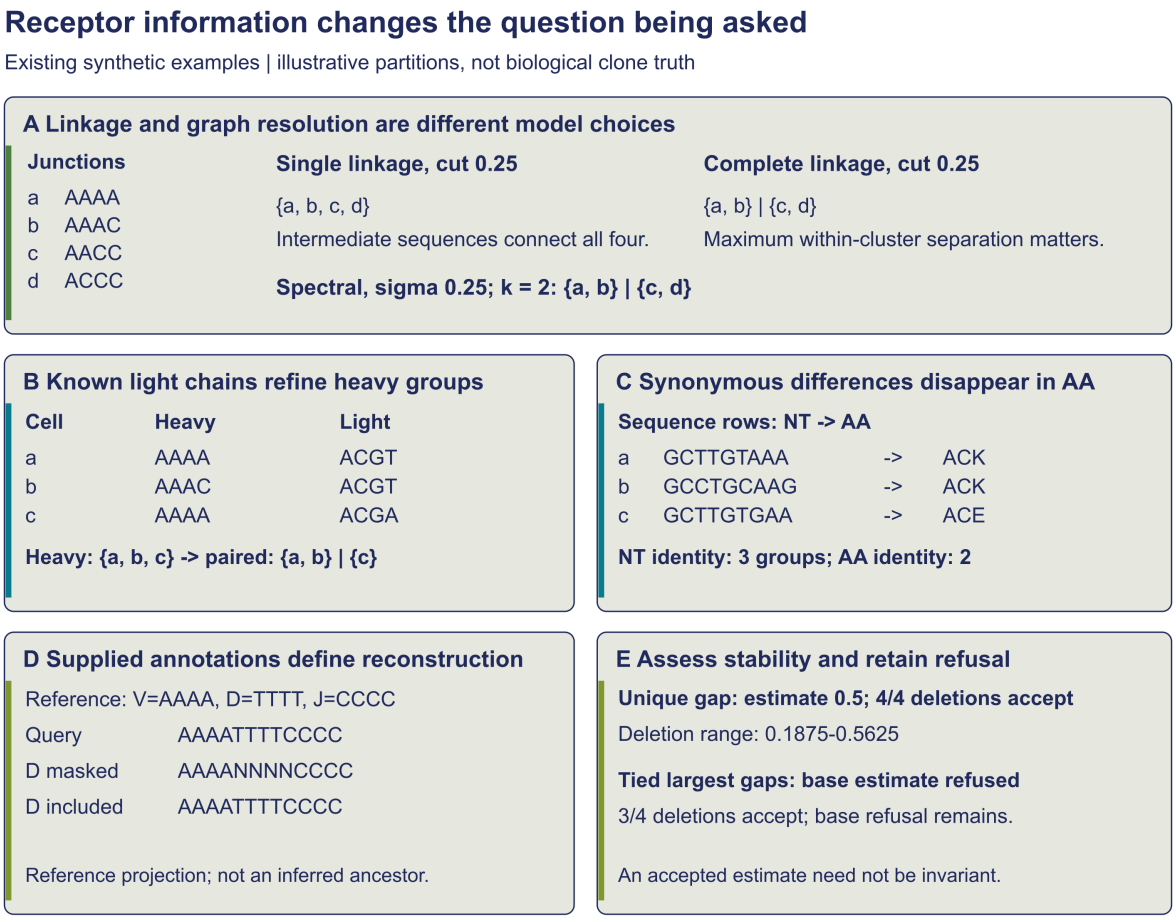


Figure 3. Extended analyses preserve distinct receptor information. Synthetic examples illustrate analysis choices. A: Single linkage at 0.25 joins AAAA, AAAC, AACC and ACCC; complete linkage at 0.25 and spectral analysis (sigma 0.25, k=2) give {a,b}|{c,d}. B: Exact known light signatures refine one heavy-chain group into {a,b}|{c}; the missing-light cell is excluded. C: Three nucleotide junctions encode ACK, ACK and ACE, giving three nucleotide-identity groups and two amino-acid-identity groups. D: Supplied V/D/J annotations reconstruct AAAANNNNCCCC with D masking or AAAATTTTCCCC with D inclusion. E: The unique-gap estimate is 0.5; all four deletion estimates are available but span 0.1875-0.5625. The tied-gap example refuses the base estimate despite three available deletion estimates. These examples do not supply biological clone truth, inferred ancestors or calibrated uncertainty. Parameters and exact inputs are in Supplementary Methods.

The contrast concerns analytical model choice, not demonstrated evolutionary intermediates or distinct biological lineages. Weighted hierarchical merges and cuts were compared with mathematical expectations, and spectral results with independent NumPy/LAPACK-based calculations. Supporting evidence includes 1,023 hierarchical cuts/merges, 740 accepted spectral trials and 40 numerical-refusal fixtures. On two prepared public tables, the workflows accounted for 2,246 and 985 rows, retaining 1,097 and 479 IGH observations. These results do not extend the selected SCOPer-compatible identity profile to adaptive spectral models; dense work units remain limited to 512 rows.

Paired-chain constraints and amino-acid representations preserve additional receptor information

Heavy-chain similarity does not capture all information available from paired receptors or protein sequences. ClonoTrace provides paired-chain and amino-acid workflows to retain selected additional distinctions.

In the paired example, heavy-chain similarity at threshold 0.25 grouped three cells together. Two shared the same known light-chain signature and the third differed; paired refinement returned {a,b}|{c} (Figure 3B). The implemented rule requires the same known light locus/gene/junction signature for cells to remain grouped after heavy-first refinement. The additional missing-light cell was excluded under the selected policy. This illustrates an analytical rule for retaining pairing information, not false biological aggregation or a universal requirement for clonal ancestry.

Sequence representation provides an orthogonal analytical choice. The three nucleotide junctions GCTTGTAAA, GCCTGCAAG and GCTTGTGAA encode ACK, ACK and ACE under standard genetic code 1, frame 0. Nucleotide identity yields three groups; amino-acid identity groups the two synonymous variants and retains the distinction that changes the translated sequence (Figure 3C). Thresholded amino-acid Hamming analysis is also available. This example demonstrates loss of synonymous nucleotide distinctions, not functional convergence, selection or neutral drift.

Reference-directed germline reconstruction and threshold sensitivity reveal supporting analytical dependencies

Reference annotations and threshold choice introduce supporting dependencies alongside the clustering model. Germline projection and deletion-based threshold assessment make those dependencies visible without reconstructing a phylogenetic lineage.

In the germline example, supplied segments V=AAAA, D=TTTT and J=CCCC and query AAAATTTTCCCC yield AAAANNNNCCCC with D masking or AAAATTTTCCCC with D inclusion (Figure 3D). Reconstruction is a reference-directed projection through supplied annotations, not ancestral inference. Public D-inclusive evidence remains narrow: three retained IGH cases in the first public dataset and none in the second. These examples do not establish broad biological reconstruction accuracy.

Threshold sensitivity analysis exposes a different dependency. In the unique-gap example, the estimate was 0.5 and all four deletion estimates were available, yet their range was 0.1875-0.5625. In the tied-gap example, equal largest gaps of 0.125 led to refusal of the base estimate despite three available deletion estimates (Figure 3E). The distinct-junction limit remains 128. Available deletion estimates can reveal sensitivity or coexist with a base refusal; they do not replace a refused base

estimate. Deletion ranges measure sensitivity, not confidence intervals or calibrated biological error rates.

Algorithmic optimization accelerates dense spectral clustering without altering numerical partitions

Dense spectral analysis involves eigendecomposition of a normalized Laplacian. Storage and traversal changes were evaluated as a way to reduce execution time while retaining the same numerical analysis.

For the 512-row sequence workflow with sigma 0.25 and eight requested clusters, the one-worker median fell from 5.029620 to 1.145595 seconds: 5.03 to 1.15 seconds, or 4.39-fold faster (Figure 2C). The five-run ranges were 4.360278-7.732536 and 1.133138-1.158013 seconds. Median maximum-process RSS increased from 35.03 to 38.09 MiB.

Existing differential comparisons comprised 1,381 cases: 1,323 accepted cases and 58 matching refusals/errors. The storage and traversal change preserved numerical results and failure behavior. At eight requested workers, the secondary before/after medians were 6.247161 and 1.127788 seconds, or 5.54-fold faster. Dense execution showed no meaningful worker scaling; each component solver remains scalar and multithreading did not cause the improvement. These measurements are specific to the recorded macOS ARM64 host and workload, with a 512-row dense work-unit limit.

Discussion

Principal contribution

ClonoTrace establishes a unified, deterministic framework that recasts B-cell receptor clonal inference as an explicit, auditable exploration of alternative receptor models. An inferred clone is not an empirical biological fact, but an analytical hypothesis defined by the sequence features, distance metrics, and clustering models applied. By unifying established reference workflows and alternative mathematical formulations within a single, reproducible architecture, ClonoTrace enables investigators to evaluate how specific analytical assumptions govern inferred clonal boundaries.

Scientific meaning of alternative receptor models

Alternative receptor models retain different information under explicit rules. Single linkage permits chains of pairwise similarity; complete linkage and spectral clustering can produce separate analytical groups under their specified models. Known light-chain signatures can refine heavy-chain groups, while amino-acid representation joins synonymous nucleotide variants and retains translated differences. Reference-directed germline projection and threshold deletion analysis expose annotation policies and sensitivity. None of these contrasts identifies the biological mechanism that generated the sequences.

Compatibility evidence

Software concordance addresses continuity with selected reference-compatible profiles. Complete member partitions matched Change-O 1.3.5 in 37/37 eligible cases and SCOPer 1.6.0 in 1,006/1,006; the descriptive count comparisons had Pearson r=1 and Spearman rho=1. The separate count and partition tests support reproducibility within those profiles, without conferring biological accuracy on their assignments.

Practical significance

For fixed inputs and configuration, deterministic execution and explicit analysis choices provide a practical basis for reproducible receptor analysis. Source-row accounting and supporting diagnostics retain the connection between data and results. The dense optimization makes the same 512-row spectral workflow 4.39-fold faster in the principal one-worker comparison, without changing its interpretation or demonstrating worker scaling.

Consolidated limitations

Equal clone counts do not establish equal member partitions; complete reference-partition agreement does not establish ancestry, biological accuracy or superiority. Designed examples illustrate analysis choices, not biological lineage truth, population generalization or calibrated sensitivity and specificity. Pearson r and Spearman rho summarize stored execution cases, including existing reruns, not independent biological samples or significance tests.

Explicit spectral k is a requested resolution, not discovery of the true number of clones or agreement with SCOPer's adaptive spectral models. Exact known light-signature refinement is a selected heavy-first model; missing or conflicting chains require explicit policies and do not establish universal biological requirements. Translation removes synonymous nucleotide distinctions and retains translated differences; the example does not establish biological function or selection. Reconstruction depends on supplied references and annotations and is not ancestral inference; public D-inclusive evidence remains narrow. Deletion ranges are sensitivity summaries, not confidence intervals or calibrated biological error rates; available replicates do not override a refused base estimate.

Measurements apply to the recorded macOS ARM64 Mac16,11 host with 24 GiB RAM and workload. Dense execution showed no meaningful worker scaling; multithreading did not cause the improvement, and RSS is not a hard RAM guarantee. Dense work units remain limited to 512 rows.

Closing perspective

Clonal inference is an explicit modeling choice governed by the question and receptor information available. By combining selected reference-compatible profiles with alternative receptor models, ClonoTrace provides an auditable basis for comparing those choices. The framework supports future evaluation against suitable independent biological evidence without treating the present computational examples as biological validation.

Software and data availability

The source release version is ClonoTrace 1.0.0. The source release includes the ClonoTrace software, documentation, and manuscript. Detailed qualification records and supporting analysis data are preserved separately. The manuscript is supplied as a standalone PDF with its figures and Supplementary Methods. Original ClonoTrace code is licensed Apache-2.0 OR MIT; third-party assets retain their own terms. Biological input datasets, biological germline databases and upstream runtime environments are not bundled.

Project repository: https://github.com/Yoonecorn/ClonoTrace. The repository contains the source release; no public-release status is asserted here.

The public-table sources were 10x Genomics, Human B cells from a Healthy Donor, 1k cells (Cell Ranger 6.0.0), and 10k Human PBMCs, 5-prime v2.0, Chromium X with intronic reads (Cell Ranger 6.1.2). Original attribution, dataset links, input checksums and preparation records are preserved with the supporting data.

References

1. Gupta NT, Vander Heiden JA, Uduman M, Gadala-Maria D, Yaari G, Kleinstein SH. Change-O: a toolkit for analyzing large-scale B cell immunoglobulin repertoire sequencing data. *Bioinformatics*. 2015;31:3356-3358. doi:10.1093/bioinformatics/btv359.

2. Nouri N, Kleinstein SH. A spectral clustering-based method for identifying clones from high-throughput B cell repertoire sequencing data. *Bioinformatics*. 2018;34:i341-i349. doi:10.1093/bioinformatics/bty235.

3. Ralph DK, Matsen FA IV. Inference of B cell clonal families using heavy/light chain pairing information. *PLoS Computational Biology*. 2022;18:e1010723. doi:10.1371/journal.pcbi.1010723.

4. Vander Heiden JA, Marquez S, Marthandan N, et al. AIRR Community Standardized Representations for Annotated Immune Repertoires. *Frontiers in Immunology*. 2018;9:2206. doi:10.3389/fimmu.2018.02206.

Supplementary Methods: reproducibility and evidence boundaries

ClonoTrace 1.0.0-rc.1. This supplement supports the revised manuscript; all scientific implementation and limits are unchanged.

S1. Native profiles and formal concordance

Change-O 1.3.5 `DefineClones` is restricted to gene resolution, normalized nucleotide Hamming distance, average symmetrization, single linkage and maxmiss=0. Thresholds are explicit and inclusive; set/first call handling follows the frozen native profile. The SCOPer 1.6.0 `identicalClones` profile uses nucleotide identity, cdr3=FALSE, mod3=FALSE and max_n=0, with the specified first/set call action. These profiles do not cover every mode of either package.

Reference observations were obtained from independently authored inputs and unmodified upstream programs run as separate processes. Full member partitions, independent of numeric labels, were the formal concordance outcome. Original inputs, outputs, checksums, options and provenance are retained in the separately preserved validation corpus. The preceding manuscript revision replayed eligible inputs with ClonoTrace 1.0.0-rc.1. This figure rearrangement reuses those results without new inference runs, native observations or timing benchmarks.

S2. Figure 2A-B eligibility and statistical unit

The frozen source containers are `changeo.json.gz` (41 records) and `scoper.json.gz` (1,006 records). A plotted observation requires a unique stored case ID, one elementary input comparison, complete observed reference clone membership, and one successful ClonoTrace replay with all rows accounted for. Reference clone counts are independently parsed from the observed output table and checked against its stored canonical partition. ClonoTrace clone counts are parsed from the replayed assignment table; complete partitions must agree before a case can be plotted.

Four Change-O records bundle independently indexed subprobes: `changeo/exhaustive-vj-sets` and `changeo/final-round-1`, `-2`, `-3`. Their aggregate partitions are present in the corpus, but they do not supply a single elementary paired clone count for this plot. They are excluded intact. No subprobe is split out, no aggregate sum is plotted and no new observation is created by merging data. The remaining 37 Change-O records and all 1,006 SCOPer records are retained. Exact exclusions and source indices are recorded in the separately preserved concordance data.

The statistical unit is a stored execution case. The Change-O subset has 23 unique input hashes and 37 distinct input/action/threshold combinations. The SCOPer subset has 799 unique input hashes and 990 distinct input/action/threshold combinations. Genuine separately stored reruns or settings are preserved as existing observations; they are not newly duplicated and are not independent biological samples. Current worker replays are never appended as additional plotted cases. Public benchmark observations are not appended to the plot.

The separately preserved concordance table gives every eligible case exactly once, including both counts, reference version, options, source/input/output hashes and partition outcome. Recorded overlap multiplicities determine marker area; a visual marker groups identical coordinates without changing the statistical rows. Pearson r is calculated on all individual paired counts. Spearman rho is Pearson correlation of average ranks, with tied values receiving midranks. No significance tests or biological independence assumptions are introduced. Both coefficients equal 1.0, and exact count

matches are 37/37 and 1,006/1,006. Each panel has square plotting geometry and equal x/y units; the panels have different count ranges for readability.

Clone-count correlation cannot establish partition equality: distinct assignments may have the same count. Formal complete-partition comparison passed for all eligible cases. Neither count correlation nor partition agreement establishes biological ancestry, biological accuracy or superiority.

S3. Extended illustration sources and parameters

Figure 3 uses existing independently authored synthetic fixtures and worked examples. The separately preserved example data record every input, expected grouping, parameter, source case and source hash. These records are separate from the frozen native correlation data.

Panel A uses the chain example from the hierarchical and spectral test set: AAAA, AAAC, AACC, ACCC. At threshold 0.25, single linkage gives one group, while complete linkage gives {a,b}|{c,d}. Spectral analysis uses sigma 0.25, k=2 and no affinity cutoff, giving {a,b}|{c,d}. The row labels a-d are display labels for the fixture's ordered rows. Average linkage is supported but is not needed to illustrate this contrast. The completed-method mathematical evidence contains 1,023 hierarchical cuts/merges, 740 accepted spectral trials and 40 numerical-refusal fixtures.

Panel B uses the worked paired-chain example. Heavy-chain threshold is 0.25, missing-light policy is `exclude` and multi-chain policy is `collapse-identical`. Three complete cells are illustrated; the additional heavy-only cell d is excluded. Refinement uses an exact known light locus/gene/junction signature, not an inferred light ancestor. Identical signatures can be collapsed computationally without losing source rows.

Panel C uses the existing three-sequence translation test example. GCTTGTAAA, GCCTGCAAG and GCTTGTGAA translate to ACK, ACK and ACE with standard genetic code 1, frame 0. Nucleotide identity here means zero-distance normalized Hamming assignment, not the method-specific SCOPer ambiguity model. AA identity joins the synonymous pair. Stops, incomplete codons and unsupported symbols remain excluded under the fixed policies. Optional IUPAC resolution accepts only unanimously translated codons.

Panel D uses the worked germline example and its independently authored synthetic references: V=AAAA, D=TTTT, J=CCCC. CIGARs are V `4M8S`, D `4S4M4S`, J `8S4M`; the query is AAAATTTTCCCC. D masking gives AAAANNNNCCCC and inclusion gives AAAATTTTCCCC. More complex independent fixtures cover insertions and deletions. Public D-inclusive evidence remains narrow (three retained IGH cases in the first public dataset and none in the second), so the illustration does not establish broad biological reconstruction accuracy.

Panel E uses the known-gap and tied-gap threshold examples. The first selects 0.5, with four accepted deletion estimates spanning 0.1875-0.5625; all replicates being available does not make the estimate invariant. The tied-gap case has equal largest gaps of 0.125 and refuses the base estimate, despite three accepted deletion estimates. A deletion estimate is not a replacement for a refused base estimate. This panel shows stability assessment versus refusal without fabricating a numerically invariant example. The absolute distinct-junction limit remains 128.

S4. Numerical and execution details

Extended bulk work units use exact donor/sample/locus/single V gene/single J gene/analyzed length. Hierarchy is row-weighted for average linkage. Spectral analysis uses a zero-diagonal Gaussian affinity, positive-edge components, a normalized Laplacian, deterministic cyclic Jacobi decomposition, canonical repeated-eigenspace bases and deterministic Lloyd assignment. The fixed limits include 100 Jacobi sweeps, off-diagonal tolerance 1e-13, eigen residual bound 1e-9, and at most 512 rows per dense work unit. No numerical rule was adjusted for this revision. The dense optimization changes workspace storage and traversal while preserving pivot order and scalar arithmetic; 1,381 differential cases comprised 1,323 accepted cases and 58 matching refusals/errors in the existing evidence.

Original rows and annotations remain accounted for. Structural, resource and numerical failures retain their existing behavior. Input hashing, atomic exclusive result publication, row accounting, schema names, deterministic labels, limits and durability semantics are specified in the accompanying user manual, native contract, completed-method contract and technical CLI/API reference (see Software and data availability). These details were removed from the main narrative rather than changed.

S5. Historical performance evidence

Native-compatible partitions and command runtimes

On the same prepared 2,246-row public B-cell receptor table, both tools retained 1,097 IGH rows, excluded 1,149 other rows and failed zero rows. Change-O-compatible assignment produced 1,092 clones and SCOPer-compatible identity produced 1,095, each matching its own complete member partition. One-worker median command times were 0.081 versus 3.223 seconds for the Change-O comparison and 0.085 versus 3.855 seconds for SCOPer. The timing differences include interpreter startup and unequal auxiliary output work, so these small-input measurements do not establish general speed ratios or parallel acceleration.

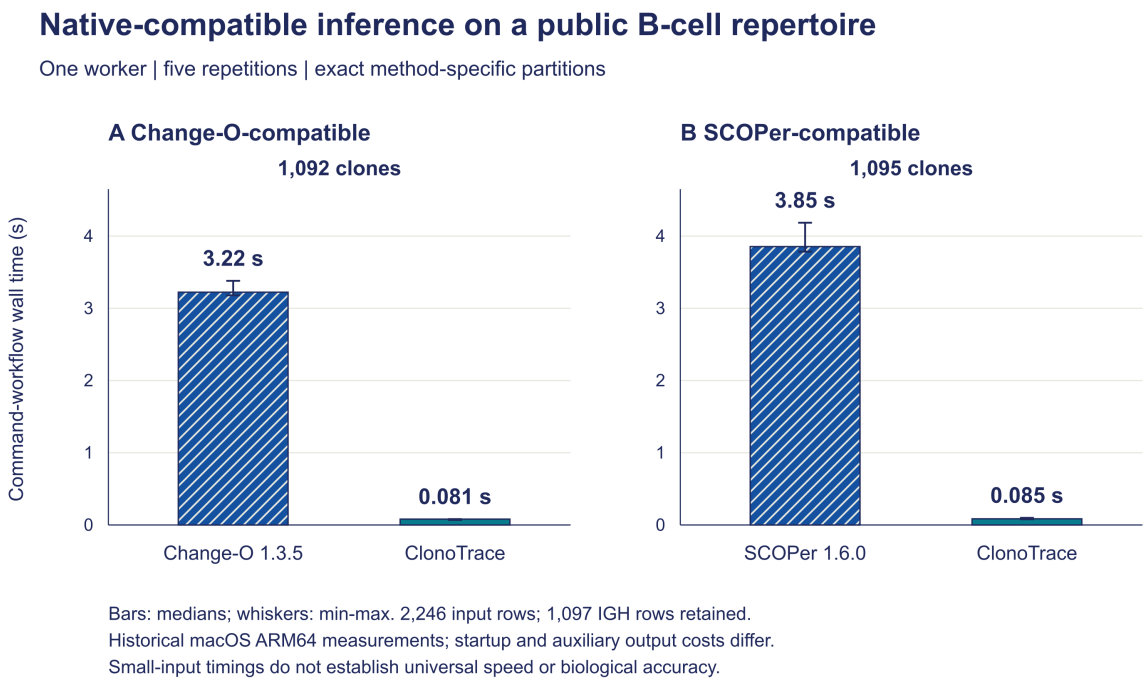


Figure S1. Native-compatible partitions and historical command runtimes. On the prepared 2,246-row public B-cell receptor table, each tool retained 1,097 IGH rows, excluded 1,149 other rows and failed zero rows. Change-O-compatible assignment at normalized Hamming threshold 0.15 gave 1,092 clones; SCOPer-compatible nucleotide identity gave 1,095. Set-call handling was used, and ClonoTrace matched each reference's complete member partition. Bars show one-worker median wall time across five repetitions; whiskers span minima and maxima. ClonoTrace/Change-O times were 0.081/3.223 s; ClonoTrace/SCOPer times were 0.085/3.855 s. Measurements include upstream interpreter startup and unequal auxiliary output costs. They are historical workflow measurements, not general speed ratios.

Dense spectral optimization

Figure 2C uses the existing 512-row synthetic sequence workflow with sigma 0.25 and eight requested clusters. The principal one-worker comparison is 5.029620 to 1.145595 s (4.39-fold); five-run ranges are 4.360278-7.732536 and 1.133138-1.158013 s. Median maximum-process RSS increased from 35.03 to 38.09 MiB. At eight requested workers, the before/after medians were 6.247161 and 1.127788 seconds, a secondary 5.54-fold improvement. Optimized one- and eight-worker times were similar: dense execution showed no meaningful worker scaling, each dense component solver remains scalar, and multithreading did not cause the improvement.

Both performance comparisons used five valid sequential repetitions, warm filesystem caches, no affinity and the recorded macOS ARM64 Mac16,11 host with 24 GiB RAM. Full-command wall time includes process startup; builds were excluded. The native wrapper includes preflight, scope projection, upstream execution and assignment emission; auxiliary outputs differ from ClonoTrace. Whiskers show observed ranges, not biological variation or inferential uncertainty. Darwin maximum-process RSS is not summed process-tree memory. Exact measurements and source hashes are retained in the separately preserved publication data. No benchmark was rerun and the one-worker 4.39-fold claim remains principal.